

Installing a guest OS

To begin OS installation, we need to attach an ISO image to the VM. Follow the steps given below to begin OS installation:

- 1) Select the newly created VM.
- 2) Click the 'Settings' button on the toolbar.
- 3) Select the storage option from the left pane.
- 4) Select the optical disk drive from the storage devices.
- 5) Provide the path of the ISO image and click the 'OK' button. Figure 3 depicts the first five steps.
- 6) Select the VM from the left pane. Click the 'Start' button on the toolbar. Follow the on-screen instructions to complete OS installation.

VM power actions

Let us understand VM power actions in detail.

- 1) **Power On:** As the name suggests, this starts the VM at the state it was powered off or saved in. To start the VM, right-click on it and select the 'Start' option.
- 2) **Pause:** In this state, the guest releases the CPU but not the memory. As a result, the contents of the memory are preserved when the VM is resumed. To pause the VM, right-click on it and select the 'Pause' option.
- 3) **Save:** This action saves the current VM state and releases the CPU as well as the memory. The saved machine can be started again in the same state. To save the VM, right-click on it and select the 'Close->Save State' option.
- 4) **Shutdown:** This is a graceful turn-off operation. In this case, the shutdown signal is sent to the guest. To shut down the VM, right-click on it and select the 'Close->ACPI Shutdown' option.
- 5) **Poweroff:** This is non-graceful turn-off operation. It can cause data loss. To power off the VM, right-click on it and select the 'Close->Poweroff' option.
- 6) **Reset:** The Reset option will turn off and turn on the VM, respectively. It is different from Restart, which is a graceful turn-off operation. To reset the VM, right-click on it and select the 'Reset' option.

Removing the VM

Let us explore the steps we need to take to remove a VM. The remove operation can be broken up into two parts.

- 1) **Unregister VM:** This removes the VM from the library, i.e., it will just unregister the VM from VirtualBox so that it won't be visible in VirtualBox Manager. To unregister a VM, right-click on it, select the 'Remove' option and click the 'Remove Only' option. You can re-register this VM by navigating to the 'Machine->Add' option from VirtualBox Manager.
- 2) **Delete VM:** This action is used to delete the VM permanently. It will delete the VM's configuration files and virtual hard disks. Once performed, this action cannot be undone. To remove a VM permanently, right-click on it, select the 'Remove' option and click the 'Delete all files' option.



Figure 2: Creating a VM

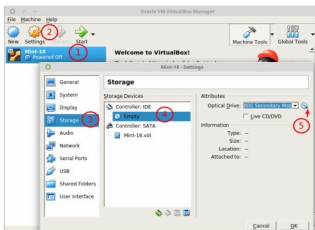


Figure 3: Installing the OS



Figure 4: Starting the VM

VirtualBox—beyond the basics

Beginners will get a fair idea about virtualisation and VirtualBox by referring to the first few sections of this article. However, VirtualBox is a feature-rich product; this section describes its more advanced features.